

# The measurable Carpenter selector problem was answered affirmatively in arXiv:1803.03632

Literature-resolution packet for arXiv:1302.6632

11 August 2026

**Status: literature already answered.** Bownik and Szyszkowski prove the measurable projection selector requested in Problem 2 of Jasper–Bownik and, as an application, the exact converse spectral-function characterization requested in Problem 1. The answer is affirmative in the source’s full finite or  $\sigma$ -finite setting, and the later projection theorem is stated for an arbitrary measurable space.

## 1 The two source questions

Let  $X$  be a finite or  $\sigma$ -finite measure space, let  $I$  be countable, and let  $\sigma : X \times I \rightarrow [0, 1]$  be measurable. For each  $\xi$ , the source forms the Kadison sums

$$a(\xi) = \sum_{\sigma(\xi, i) < 1/2} \sigma(\xi, i), \quad b(\xi) = \sum_{\sigma(\xi, i) \geq 1/2} (1 - \sigma(\xi, i)).$$

Assume pointwise almost everywhere that either one sum is infinite, or both are finite and  $a - b \in \mathbb{Z}$ .

Problem 2 asks whether there is a measurable range function  $J(\xi)$  whose orthogonal projection has diagonal  $(\sigma(\xi, i))_{i \in I}$ . Problem 1 is the shift-invariant specialization: do these pointwise conditions exactly characterize spectral functions of shift-invariant subspaces of  $L^2(\mathbb{R}^d)$ ? The source explicitly says the measurable-selector issue is the missing step. Its complete question page is reproduced next.

which identifies range functions if they are equal a.e. A fundamental result due to Helson [12, Theorem 8, p. 59] gives one-to-one correspondence between SI spaces  $V$  and measurable range functions  $J$ , see also [7, Proposition 1.5]. Among several equivalent ways of introducing the spectral function of a SI space the most relevant definition uses a range function.

**Definition 5.1.** The *spectral function* of a SI space  $V$  is a measurable mapping  $\sigma_V : \mathbb{R}^d \rightarrow [0, 1]$  given by

$$(5.1) \quad \sigma_V(\xi + k) = \|P_J(\xi)e_k\|^2 = \langle P_J(\xi)e_k, e_k \rangle \quad \text{for } \xi \in \mathbb{T}^d, \ k \in \mathbb{Z}^d,$$

where  $\{e_k\}_{k \in \mathbb{Z}^d}$  denotes the standard basis of  $\ell^2(\mathbb{Z}^d)$  and  $\mathbb{T}^d = [-1/2, 1/2)^d$ . In other words,  $\{\sigma_V(\xi + k)\}_{k \in \mathbb{Z}^d}$  is a diagonal of a projection  $P_J(\xi)$ .

Note that  $\sigma_V(\xi)$  is well defined for a.e.  $\xi \in \mathbb{R}^d$ , since  $\{k + \mathbb{T}^d : k \in \mathbb{Z}^d\}$  is a partition of  $\mathbb{R}^d$ . As an immediate consequence of Theorem 1.1 we have the following result.

**Theorem 5.2.** Suppose that  $V \subset L^2(\mathbb{R}^d)$  is a SI space. Let  $\sigma = \sigma_V : \mathbb{R}^d \rightarrow [0, 1]$  be its spectral function. For  $\xi \in \mathbb{T}^d$  define

$$a(\xi) = \sum_{k \in \mathbb{Z}^d, \sigma(\xi+k) < 1/2} \sigma(\xi+k) \quad \text{and} \quad b(\xi) = \sum_{k \in \mathbb{Z}^d, \sigma(\xi+k) \geq 1/2} (1 - \sigma(\xi+k)).$$

Then, for a.e.  $\xi \in \mathbb{R}^d$  we either have

- (i)  $a(\xi), b(\xi) < \infty$  and  $a(\xi) - b(\xi) \in \mathbb{Z}$ , or
- (ii)  $a(\xi) = \infty$  or  $b(\xi) = \infty$ .

It is an open problem whether the converse to Theorem 5.2 holds.

**Problem 1.** Suppose that a measurable function  $\sigma : \mathbb{R}^d \rightarrow [0, 1]$  satisfies either (i) or (ii) for a.e.  $\xi \in \mathbb{R}^d$ . Does there exist a SI space  $V \subset L^2(\mathbb{R}^d)$  such that its spectral function  $\sigma_V = \sigma$ ?

The sufficiency part of Theorem 1.1 i.e., Carpenter's Theorem, suggests a positive answer to this problem. That is, for a.e.  $\xi$  it yields a projection  $P_J(\xi)$  whose diagonal satisfies (5.1). However, it does not guarantee a priori that the corresponding range function  $J$  is measurable. This naturally leads to the following selector problem.

**Problem 2.** Let  $X$  be a finite (or  $\sigma$ -finite) measure space and let  $I$  be a countable index set. Let  $\sigma : X \times I \rightarrow [0, 1]$  be a measurable function. For  $\xi \in X$  define

$$a(\xi) = \sum_{i \in I, \sigma(\xi, i) < 1/2} \sigma(\xi, i) \quad \text{and} \quad b(\xi) = \sum_{i \in I, \sigma(\xi, i) \geq 1/2} (1 - \sigma(\xi, i)).$$

Suppose that for a.e.  $\xi \in X$  we either have

- (i)  $a(\xi), b(\xi) < \infty$  and  $a(\xi) - b(\xi) \in \mathbb{Z}$ , or
- (ii)  $a(\xi) = \infty$  or  $b(\xi) = \infty$ .

Does there exist a measurable range function  $J : X \rightarrow \{\text{closed subspaces of } \ell^2(I)\}$  such that the corresponding orthogonal projections  $P_J(\xi)$  have diagonal  $\{\sigma(\xi, i)\}_{i \in I}$  for a.e.  $\xi \in X$ ?

In other words, Problem 2 asks whether it is possible to find a measurable selector of projections in Theorem 1.1. The constructive proof of Carpenter's Theorem given in this paper might be a first step toward resolving this problem. However, Problem 2 remains open.

## 2 The exact later theorem

The abstract of Bownik–Szyszkowski’s arXiv:1803.03632 says explicitly that the paper solves the problem posed by Jasper and Bownik. Its title is *Measurable selector in Kadison’s carpenter’s theorem*. Theorem 1.2 states:

For any measurable space  $X$  and measurable functions  $f_i : X \rightarrow [0, 1]$ , if Kadison’s divergence-or-integer alternative holds at every point, there is a weak-operator measurable projection  $P : X \rightarrow \mathcal{B}(H)$  with diagonal  $(f_i)$ .

This is exactly the requested selector after identifying the countable set  $I$  with  $\mathbb{N}$  and discarding a null exceptional set (or assigning arbitrary admissible data there). The later result is slightly stronger: it imposes no finite or  $\sigma$ -finite measure hypothesis on  $X$ .

The two papers use opposite conventions for entries equal to  $1/2$ . This is harmless. Moving one such entry from the upper sum to the lower sum changes  $a - b$  by 1, preserving integrality; infinitely many such entries force the divergent alternative.

The complete theorem page follows.

operator was studied by Arveson [6], and by Kennedy and Skoufranis [37], who have shown approximate Carpenter's Theorem for the closure of  $D_{\mathcal{A}}(T)$  in von Neumann factors of type  $I_\infty$ , II, and III. In the setting of von Neumann factor  $I_\infty$ , when  $\mathfrak{M} = \mathcal{B}(H)$ , Problem 1 was studied by Kaftal, Loreaux, and Weiss [33, 34] when  $T$  is a positive compact operator and by Jasper and the first author when  $T$  has finite spectrum [17, 29].

The goal of this paper is to show the existence of a measurable selector in Kadison's Carpenter's Theorem. This problem was explicitly posed by Jasper and the first author in [16], although it was motivated by the earlier work of Rzeszotnik and the first author on spectral functions of shift-invariant spaces [18, 19].

**Definition 1.1.** Let  $X$  be a measurable space and  $H$  be a separable Hilbert space. We say that  $P : X \rightarrow \mathcal{B}(H)$  is a *measurable projection*, if:

- (i)  $P(x)$  is an orthogonal projection for all  $x \in X$ , and
- (ii)  $P$  is weakly operator measurable, i.e.,

$$x \mapsto \langle P(x)u, v \rangle \text{ is measurable for all } u, v \in H.$$

Equivalently, all entries of  $P(x)$  with respect to a fixed orthonormal basis  $(e_i)_{i \in \mathbb{N}}$  of  $H$  are measurable.

The *diagonal* of  $P$  is a sequence of measurable functions  $(\langle P(x)e_i, e_i \rangle)_{i \in \mathbb{N}}$ .

Our main result takes the following form.

**Theorem 1.2.** Let  $X$  be a measurable space. Let  $f_i : X \rightarrow [0, 1]$ ,  $i \in \mathbb{N}$ , be measurable functions. Define functions  $a, b : X \rightarrow [0, \infty]$  by

$$(1.2) \quad a(x) = \sum_{i \in \mathbb{N}: f_i(x) \leq \frac{1}{2}} f_i(x) \quad \text{and} \quad b(x) = \sum_{i \in \mathbb{N}: f_i(x) > \frac{1}{2}} 1 - f_i(x) \quad \text{for } x \in X.$$

Assume that for every  $x \in X$ , we have either:

- (i)  $a(x) = \infty$  or  $b(x) = \infty$ , or
- (ii)  $a(x), b(x) < \infty$  and  $a(x) - b(x) \in \mathbb{Z}$ .

Then, there exists a measurable projection  $P : X \rightarrow \mathcal{B}(H)$  with diagonal  $(f_i)_{i \in \mathbb{N}}$ .

Theorem 1.2 also yields a characterization of spectral functions of shift-invariant subspaces of  $L^2(\mathbb{R}^n)$ , which were introduced and studied by Rzeszotnik and the first author [18, 19]. In fact, this was the main motivation for studying this problem. In addition, Theorem 1.2 solves Problem 1 for von Neumann algebras of type  $I_\infty$  when  $T$  is a projection.

The proof of Theorem 1.2 splits into two natural cases: the nonsummable case (i) and summable case (ii), which are shown in Sections 2 and 3 respectively. The nonsummable case of Theorem 1.2 is based on an algorithmic technique for finding a projection with all diagonal entries in  $[0, 1/2]$  except possibly one term. This technique was introduced by Jasper and the first author in [16, Section 4]. It is related to the spectral tetris construction of tight frames introduced by Casazza, Fickus, Mixon, Wang, and Zhou in [21]. The proof of the case (i) also relies heavily on techniques of measurable permutations.

The proof of the summable case (ii) employs a measurable variant of the finite dimensional Schur-Horn theorem which was shown by Benac, Massey, and Stojanoff in [8]. The key role is played by a decoupling procedure that splits a desired diagonal sequence into three parts modifying at most one entry in each group. The resulting sequences correspond either

### 3 The shift-invariant consequence

Theorem 4.1 of arXiv:1803.03632 applies the measurable selector to the fibers  $(\sigma(\xi + k))_{k \in \mathbb{Z}^d}$ . It proves the exact equivalence:

1.  $\sigma$  is almost everywhere the spectral function of a shift-invariant subspace of  $L^2(\mathbb{R}^d)$ ;
2. Kadison's divergence-or-integer condition holds almost everywhere.

The proof explains the precise bridge: Theorem 1.2 produces the measurable field of projections, hence a measurable range function, and Helson's theorem then produces the shift-invariant space. Thus both source problems are answered affirmatively, not merely a special case.

**Theorem 4.1.** *Let  $\sigma : \mathbb{R}^d \rightarrow [0, 1]$  be a measurable function. For  $\xi \in \mathbb{R}^d$  define*

$$a(\xi) = \sum_{k \in \mathbb{Z}^d, \sigma(\xi+k) < 1/2} \sigma(\xi+k) \quad \text{and} \quad b(\xi) = \sum_{k \in \mathbb{Z}^d, \sigma(\xi+k) \geq 1/2} (1 - \sigma(\xi+k)).$$

*The following are equivalent:*

- (i) *There exists a SI space  $V \subset L^2(\mathbb{R}^d)$  such that its spectral function  $\sigma_V$  coincides almost everywhere with  $\sigma$ ,*  

$$\sigma_V(\xi) = \sigma(\xi) \quad \text{for a.e. } \xi \in \mathbb{R}^d.$$
- (ii) *For a.e.  $\xi \in \mathbb{R}^d$  we either have*
  - $a(\xi), b(\xi) < \infty$  and  $a(\xi) - b(\xi) \in \mathbb{Z}$ , or
  - $a(\xi) = \infty$  or  $b(\xi) = \infty$ .

*Proof.* The implication (i)  $\implies$  (ii) follows from the necessity part of Kadison Theorem 1.1. Indeed, by Definition 1.1 the sequence  $(\sigma_V(\xi + k))_{k \in \mathbb{Z}^d}$  is a diagonal of a projection  $P_J(\xi)$  acting on  $\ell^2(\mathbb{Z}^d)$  for a.e.  $\xi \in \mathbb{T}^d$ . Hence, (ii) holds pointwise a.e.

The converse implication (ii)  $\implies$  (i) is the crux of this result. Kadison's Theorem 1.1 implies merely the existence of projections  $P(\xi)$  with desired diagonal  $(\sigma(\xi + k))_{k \in \mathbb{Z}^d}$  for a.e.  $\xi$ . However, it does not guarantee the measurability of  $\xi \mapsto P(\xi)$ . This is where we need to apply Theorem 1.2 instead. Let  $X$  be the set of all points  $\xi \in \mathbb{R}^d$  such that (ii) holds. This is a set of full measure, i.e.,  $\mathbb{R}^d \setminus X$  is a null set. Theorem 1.2 yields the existence of a measurable projection  $P : X \rightarrow \mathcal{B}(\ell^2(\mathbb{Z}^d))$  such that  $P(\xi)$  has diagonal  $(\sigma(\xi + k))_{k \in \mathbb{Z}^d}$  for a.e.  $\xi$ . This projection correspond to a measurable range function  $J$  as in (4.1), which by Helson's theorem corresponds to a SI space  $V \subset L^2(\mathbb{R}^d)$ .  $\square$

The notion of a spectral function can be extended to the setting of translation invariant (TI) subspaces of  $L^2(G)$  of a second countable locally countable group  $G$ , see [20]. In this case,  $\Gamma \subset G$  is a closed co-compact subgroup and the spectral function represents diagonal entries of a measurable projection defined on  $X = \hat{G}/\Gamma^*$  with values in  $\mathcal{B}(\ell^2(\Gamma^*))$ , where  $\Gamma^*$  is the annihilator of  $\Gamma$  in the dual group  $\hat{G}$ . Since Theorem 1.2 does not put any restriction on a measurable space  $X$ , an extension of Theorem 4.1 to this setting does not cause any extra difficulties.

**4.2. Carpenter's Theorem for type  $I_\infty$  von Neumann algebras.** In this subsection we will answer Problem 1 for von Neumann algebras of type  $I_\infty$  when  $T$  is a projection.

The structure theorem for type I von Neumann algebras [11] Section I.3.12 gives the following characterization. A type  $I_n$ , where  $n \in \mathbb{N} \cup \{\infty\}$ , von Neumann algebra  $\mathfrak{M}$  is isomorphic with the tensor product  $\mathfrak{M} \cong \mathcal{B}(H_n) \otimes L^\infty(X, \mu)$ , where  $H_n$  is  $n$  dimensional Hilbert space and  $(X, \mu)$  is locally finite measure space. Hence,  $\mathfrak{M}$  is isomorphic to  $L^\infty(X, \mu, \mathcal{B}(H_n))$  acting on a Hilbert space  $L^2(X, \mu, H_n)$ .

More precisely, let  $H$  be the direct integral of a measurable field of infinite dimensional separable Hilbert spaces  $H_\infty$  on  $(X, \mu)$  given by

$$(4.3) \quad H = \int_X^\oplus H_\infty d\mu(x).$$

Let  $\mathfrak{M}$  be the subalgebra of  $\mathcal{B}(H)$  consisting of all decomposable operators

$$(4.4) \quad T = \int_X^\oplus T(x) d\mu(x),$$

## 4 Conclusion and audit

The match is exact in hypotheses, object constructed, and application: Problem 2 is Theorem 1.2, and Problem 1 is Theorem 4.1 of the later paper. No independent proof or counterexample should be promoted for this target.

**Source audit.** The official arXiv PDFs of 1302.6632 and 1803.03632 were inspected. Bounded exact-title and exact-phrase searches recovered the later paper as the direct resolution. The target was absent from the run’s cheap solution indexes before this packet.

**Human review.** Confirm the harmless  $1/2$  convention change and the identification of a countable index set with  $\mathbb{N}$ ; the theorem match is otherwise literal.

## References

- [1] M. Bownik and J. Jasper, *Constructive proof of the Carpenter’s Theorem*, arXiv:1302.6632.
- [2] M. Bownik and M. Szyszkowski, *Measurable selector in Kadison’s carpenter’s theorem*, arXiv:1803.03632.